

Администрирование локальных сетей

Лабораторная работа №8

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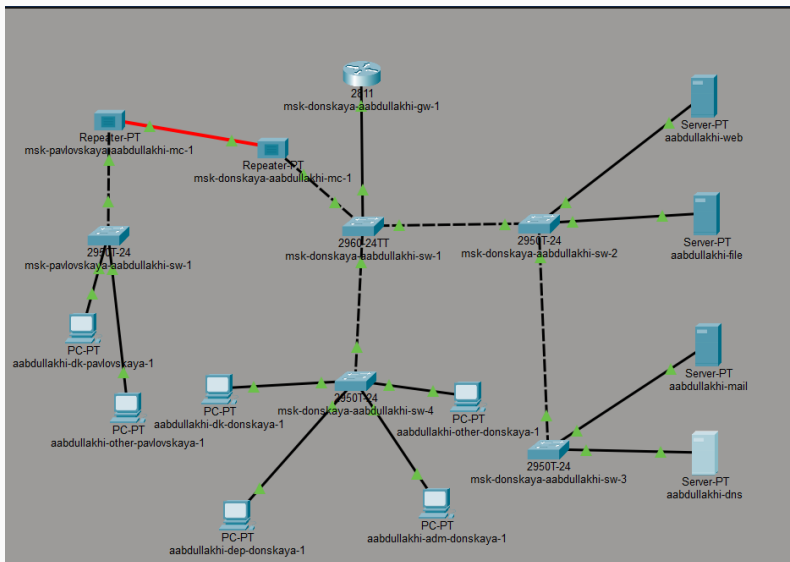
27 февраля 2026

Российский университет дружбы народов, Москва, Россия

Цель

Освоение практических навыков настройки динамического распределения IP-адресов с использованием протокола DHCP (Dynamic Host Configuration Protocol) в локальной вычислительной сети.

Выполнение лабораторной работы



Настройка IP-адреса DNS-сервера

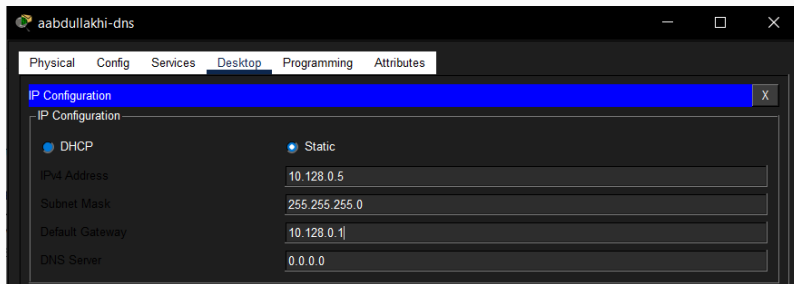


Рис. 2: Настройка IP-адреса DNS-сервера

Настройка DNS-сервиса

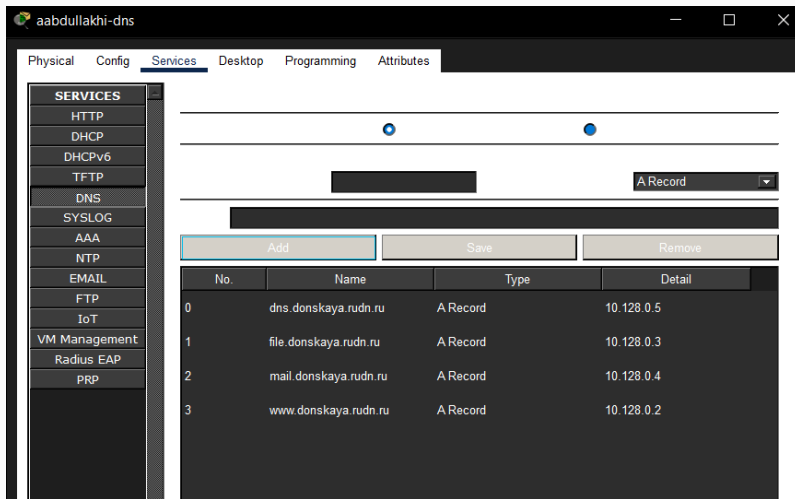
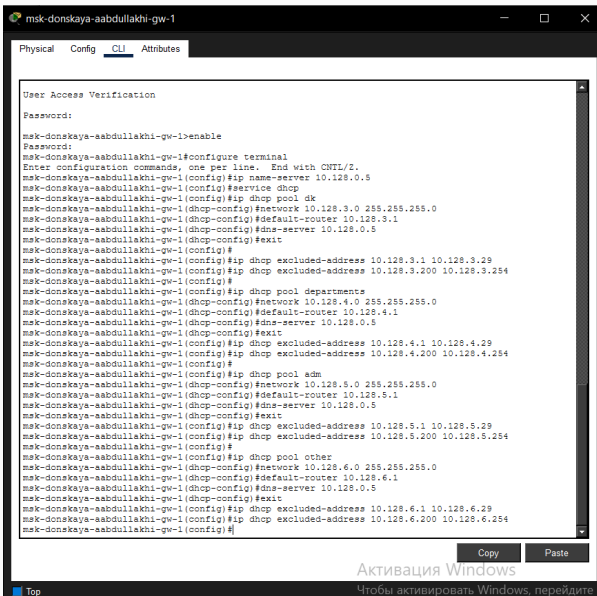


Рис. 3: Настройка DNS-сервиса

Настройка DHCP на маршрутизаторе



```
msk-donskaya-aabdullakhi-gw-1
Physical Config CLI Attributes

User Access Verification

Password:

msk-donskaya-aabdullakhi-gw-1>enable
Password:
msk-donskaya-aabdullakhi-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-aabdullakhi-gw-1 (config)#ip name-server 10.128.0.5
msk-donskaya-aabdullakhi-gw-1 (config)#service dhcp
msk-donskaya-aabdullakhi-gw-1 (config)#ip dhcp pool dk
msk-donskaya-aabdullakhi-gw-1 (dhcp-config)#network 10.128.3.0 255.255.255.0
msk-donskaya-aabdullakhi-gw-1 (dhcp-config)#default-router 10.128.3.1
msk-donskaya-aabdullakhi-gw-1 (dhcp-config)#dns-server 10.128.0.5
msk-donskaya-aabdullakhi-gw-1 (dhcp-config)#exit
msk-donskaya-aabdullakhi-gw-1 (config)#
msk-donskaya-aabdullakhi-gw-1 (config)#ip dhcp excluded-address 10.128.3.1 10.128.3.29
msk-donskaya-aabdullakhi-gw-1 (config)#ip dhcp excluded-address 10.128.3.200 10.128.3.254
msk-donskaya-aabdullakhi-gw-1 (config)#
msk-donskaya-aabdullakhi-gw-1 (config)#ip dhcp pool departments
msk-donskaya-aabdullakhi-gw-1 (dhcp-config)#network 10.128.4.0 255.255.255.0
msk-donskaya-aabdullakhi-gw-1 (dhcp-config)#default-router 10.128.4.1
msk-donskaya-aabdullakhi-gw-1 (dhcp-config)#dns-server 10.128.0.5
msk-donskaya-aabdullakhi-gw-1 (dhcp-config)#exit
msk-donskaya-aabdullakhi-gw-1 (config)#ip dhcp excluded-address 10.128.4.1 10.128.4.29
msk-donskaya-aabdullakhi-gw-1 (config)#ip dhcp excluded-address 10.128.4.200 10.128.4.254
msk-donskaya-aabdullakhi-gw-1 (config)#
msk-donskaya-aabdullakhi-gw-1 (config)#ip dhcp pool adm
msk-donskaya-aabdullakhi-gw-1 (dhcp-config)#network 10.128.5.0 255.255.255.0
msk-donskaya-aabdullakhi-gw-1 (dhcp-config)#default-router 10.128.5.1
msk-donskaya-aabdullakhi-gw-1 (dhcp-config)#dns-server 10.128.0.5
msk-donskaya-aabdullakhi-gw-1 (dhcp-config)#exit
msk-donskaya-aabdullakhi-gw-1 (config)#ip dhcp excluded-address 10.128.5.1 10.128.5.29
msk-donskaya-aabdullakhi-gw-1 (config)#ip dhcp excluded-address 10.128.5.200 10.128.5.254
msk-donskaya-aabdullakhi-gw-1 (config)#
msk-donskaya-aabdullakhi-gw-1 (config)#ip dhcp pool other
msk-donskaya-aabdullakhi-gw-1 (dhcp-config)#network 10.128.6.0 255.255.255.0
msk-donskaya-aabdullakhi-gw-1 (dhcp-config)#default-router 10.128.6.1
msk-donskaya-aabdullakhi-gw-1 (dhcp-config)#dns-server 10.128.0.5
msk-donskaya-aabdullakhi-gw-1 (dhcp-config)#exit
msk-donskaya-aabdullakhi-gw-1 (config)#ip dhcp excluded-address 10.128.6.1 10.128.6.29
msk-donskaya-aabdullakhi-gw-1 (config)#ip dhcp excluded-address 10.128.6.200 10.128.6.254
msk-donskaya-aabdullakhi-gw-1 (config)#
```

Copy Paste

Активация Windows

Чтобы активировать Windows, перейдите в

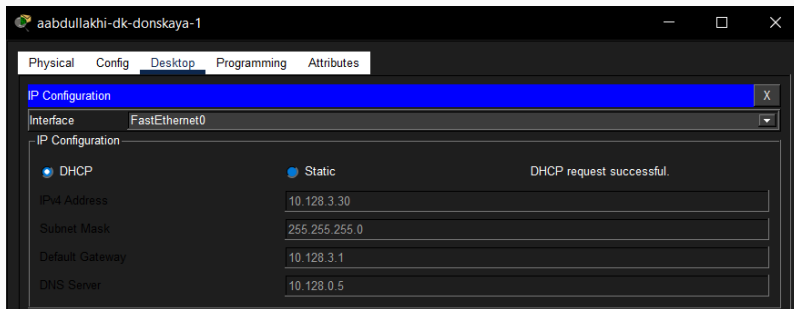
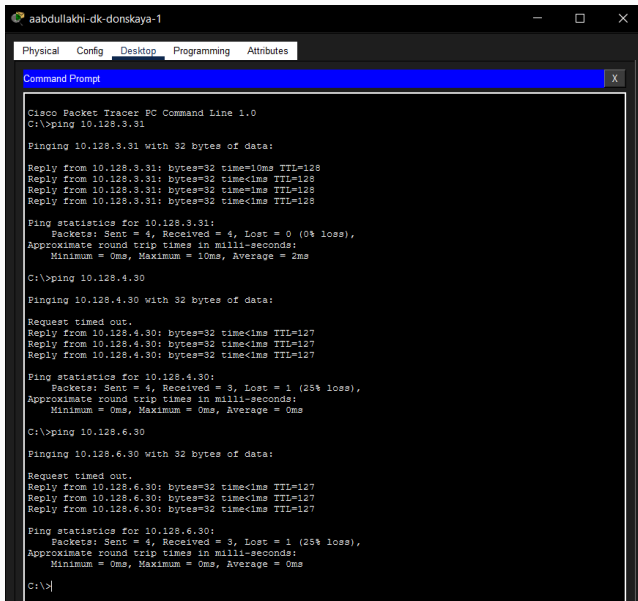


Рис. 5: получение IP-адреса по DHCP

Проверка сетевой доступности



The screenshot shows a Cisco Packet Tracer PC Command Line window for a device named 'aabdullakhi-dk-donskaya-1'. The window has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes', with 'Desktop' selected. The 'Command Prompt' window is open, displaying the results of three ping commands. The first ping to 10.128.3.31 is successful. The second ping to 10.128.4.30 shows a 25% packet loss. The third ping to 10.128.6.30 also shows a 25% packet loss.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.128.3.31

Pinging 10.128.3.31 with 32 bytes of data:

Reply from 10.128.3.31: bytes=32 time=10ms TTL=128
Reply from 10.128.3.31: bytes=32 time<1ms TTL=128
Reply from 10.128.3.31: bytes=32 time<1ms TTL=128
Reply from 10.128.3.31: bytes=32 time<1ms TTL=128

Ping statistics for 10.128.3.31:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ping 10.128.4.30

Pinging 10.128.4.30 with 32 bytes of data:

Request timed out.
Reply from 10.128.4.30: bytes=32 time<1ms TTL=127
Reply from 10.128.4.30: bytes=32 time<1ms TTL=127
Reply from 10.128.4.30: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.4.30:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.128.6.30

Pinging 10.128.6.30 with 32 bytes of data:

Request timed out.
Reply from 10.128.6.30: bytes=32 time<1ms TTL=127
Reply from 10.128.6.30: bytes=32 time<1ms TTL=127
Reply from 10.128.6.30: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.6.30:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

Информация о пулах DHCP

```
msk-donskaya-aabdullakhi-gw-1
Physical Config CLI Attributes
Leased addresses      : 2
Excluded addresses    : 8
Pending event        : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
10.128.3.1         10.128.3.1 - 10.128.3.254      2 / 8 / 254

Pool departments :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next)    : 0 / 0
Total addresses             : 254
Leased addresses            : 1
Excluded addresses          : 8
Pending event               : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
10.128.4.1         10.128.4.1 - 10.128.4.254      1 / 8 / 254

Pool adm :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next)    : 0 / 0
Total addresses             : 254
Leased addresses            : 1
Excluded addresses          : 8
Pending event               : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
10.128.5.1         10.128.5.1 - 10.128.5.254      1 / 8 / 254

Pool other :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next)    : 0 / 0
Total addresses             : 254
Leased addresses            : 2
Excluded addresses          : 8
Pending event               : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
10.128.6.1         10.128.6.1 - 10.128.6.254      2 / 8 / 254
msk-donskaya-aabdullakhi-gw-1#
```

Таблица выданных DHCP-адресов

```
msk-donskaya-aabdullakhi-gw-1#sh ip dhcp binding
IP address      Client-ID/      Lease expiration  Type
Hardware address
10.128.3.30      00D0.BAE1.7912   --                Automatic
10.128.3.31      00D0.BC50.DE5B   --                Automatic
10.128.4.30      0060.47DC.6A51   --                Automatic
10.128.5.30      0002.4A1C.C9C3   --                Automatic
10.128.6.30      000C.8566.6B15   --                Automatic
10.128.6.31      0001.9775.0E77   --                Automatic
msk-donskaya-aabdullakhi-gw-1#
```

Рис. 8: Таблица выданных DHCP-адресов

DHCP Discover

PDU Information at Device: msk-donskaya-aabdullakhi-sw-4

OSI Model Inbound PDU Details Outbound PDU Details

DATA (VARIABLE LENGTH)

DHCP

081624

Bytes

OP:0x0000000000000001

HW TYPE:1

HW LEN:6

HOPS:0

TRANSACTION ID

SECS:0

FLAGS:0x00000000000000000000000000000000

CLIENT ADDRESS:0.0.0.0

YOUR CLIENT ADDRESS:0.0.0.0|

SERVER ADDRESS:0.0.0.0

RELAY AGENT ADDRESS:0.0.0.0

CLIENT HARDWARE ADDRESS:00D0.BAE1.7912

SERVER HOSTNAME (64 BYTES)

FILE (128 BYTES)

OPTIONS (312 BYTES)

DHCP Client Identifier Option

04812

Bits

OP:0x3d

LEN:0x3d

HW:0x1

Рис. 9: DHCP Discover

DHCP Offer

PDU Information at Device: msk-donskaya-aabdullakhi-sw-1

OSI Model Inbound PDU Details Outbound PDU Details

DATA (VARIABLE LENGTH)

DHCP

081624

Bytes

OP: 0x0000000000000002

HW TYPE: 1

HW LEN: 6

HOPS: 0

TRANSACTION ID

SECS: 0

FLAGS: 0x00000000000000000000000000000000

CLIENT ADDRESS: 0.0.0.0

YOUR CLIENT ADDRESS: 10.128.3.30

SERVER ADDRESS: 10.128.3.1

RELAY AGENT ADDRESS: 0.0.0.0

CLIENT HARDWARE ADDRESS: 00D0.BAE1.7912

SERVER HOSTNAME (64 BYTES)

FILE (128 BYTES)

OPTIONS (312 BYTES)

DHCP Domain Name Server Option

048

Bits

OP: 6

LEN: 4

DOMAIN NAME SERVER: 10.128.0.5

Рис. 10: DHCP Offer

DHCP Request

PDU Information at Device: msk-donskaya-aabdullakhi-sw-4

OSI Model [Inbound PDU Details](#) [Outbound PDU Details](#)

DATA (VARIABLE LENGTH)

DHCP

0	8	16	24	Bytes
OP:0x0000000000000001	HW TYPE:1	HW LEN:6	HOPS:0	
TRANSACTION ID				
SECS:0		FLAGS:0x00000000000000000000000000000000		
CLIENT ADDRESS:0.0.0.0				
YOUR CLIENT ADDRESS:10.128.3.30				
SERVER ADDRESS:10.128.3.1				
RELAY AGENT ADDRESS:0.0.0.0				
CLIENT HARDWARE ADDRESS:00D0.BAE1.7912				
SERVER HOSTNAME (64 BYTES)				
FILE (128 BYTES)				
OPTIONS (312 BYTES)				

DHCP Client Identifier Option

0	4	8	12	Bits
OP:0x3d	LEN:0x3d	HW:0x1		
CLIENT IDENTIFIER (VARIABLE LENGTH):00D0.BAE1.7912				

DHCP Acknowledgment (ACK)

PDU Information at Device: msk-donskaya-aabdullakhi-sw-4

OSI Model

Inbound PDU Details

Outbound PDU Details

DATA (VARIABLE LENGTH)

DHCP

081624

Bytes

OP:0x0000000000000002

HW TYPE:1

HW LEN:6

HOPS:0

TRANSACTION ID

SECS:0

FLAGS:0x00000000000000000000000000000000

CLIENT ADDRESS:0.0.0.0

YOUR CLIENT ADDRESS:10.128.3.30

SERVER ADDRESS:10.128.3.1

RELAY AGENT ADDRESS:0.0.0.0

CLIENT HARDWARE ADDRESS:00D0.BAE1.7912

SERVER HOSTNAME (64 BYTES)

FILE (128 BYTES)

OPTIONS (312 BYTES)

DHCP Domain Name Server Option

048

Bits

OP:6

LEN:4

DOMAIN NAME SERVER:10.128.0.5

Процесс обмена DHCP-сообщениями в режиме симуляции

Cisco Packet Tracer 9.0.0 (saves) lab4.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical x: 579, y: 692

Simulation Panel

Event List

Vis	Time(sec)	Last Device
	0.000	--
	0.001	aabdullakhi-dk-donskaya-1
Visible	0.002	msk-donskaya-aabdullakhi-sw-4
	0.003	msk-donskaya-aabdullakhi-sw-1
	0.003	msk-donskaya-aabdullakhi-sw-1
	0.003	msk-donskaya-aabdullakhi-sw-1
	0.003	--
	0.004	msk-donskaya-aabdullakhi-mc-1
	0.004	msk-donskaya-aabdullakhi-gw-1
	0.004	msk-donskaya-aabdullakhi-sw-2
	0.005	msk-pavlovskaya-aabdullakhi-mc-1
	0.005	msk-donskaya-aabdullakhi-sw-1
	0.006	msk-pavlovskaya-aabdullakhi-sw-1
	0.006	msk-donskaya-aabdullakhi-sw-4
	0.408	--
	0.409	--

Reset Simulation ☒ Constant Delay Captured to: 1.802 s

Play Controls

Event List Filters - Visible Events

ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPsec, ISAKMP, IoT, IoT TCP, LACP, LDP, MDIOBUS, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoE, PTP, Profinet, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

Edit Filters Show All/None

Time: 00:32:00.596 PLAY CONTROLS

Scenario 0

New Delete

Toggle PDU List Window

Активация Windows
Чтобы активировать Windows, перейдите в "Параметры".

Вывод

В ходе работы был настроен DHCP-сервер на маршрутизаторе `msk-donskaya-aabdullakhi-gw-1`, созданы пулы адресов для четырёх подсетей. Оконечные устройства успешно получили IP-адреса динамически, а проверка `ping` подтвердила связность между подсетями. Процесс обмена сообщениями DHCP (Discover, Offer, Request, ACK) изучен в режиме симуляции.